



Debre Berhan University

College of Computing

Information Systems Department

Data Science Applications [InSy3056] Group Project

Ethiopian Household Wealth Prediction

A Machine Learning System for Classifying Household Wealth Quintiles

Using Five Waves of World Bank LSMS-ISA - Ethiopian Socioeconomic Survey Data

Technical and Academic Documentation

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Prepared By:

No.	Name	ID No.
1	Mesfin Maru	DBU1702015
2	Ayires Zebene	DBU1601409
3	Melkamsew Alehegn	DBU1601356
4	Kalkidan Minda	DBU1601280
5	Sofia Mohammed	DBU1501477
6	Samrawit Assefa	DBU1601585

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ABSTRACT

This document presents the complete technical and academic documentation for the Ethiopian Household Wealth Prediction System, a machine learning pipeline and interactive web application designed to classify Ethiopian households into consumption expenditure quintiles (Q1 = poorest to Q5 = wealthiest). The system integrates five waves of the World Bank Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA) and Ethiopian Socioeconomic Survey (ESS) covering the years 2011–2022.

The pipeline includes multi-format data ingestion, survey-aware missing value imputation, domain-informed feature engineering, multi-class supervised learning, regional wealth modeling, and an interactive Streamlit dashboard deployed publicly at ethiowealthpredictor.streamlit.app.

The best-performing model, a Random Forest classifier, achieved a test accuracy of 40.81%, a weighted F1-score of 0.3992, and a macro F1-score of 0.3714, with a five-fold stratified cross-validation mean F1-score of 0.3839 (± 0.0057). These results demonstrate substantial predictive power relative to a random baseline of approximately 20% for a balanced five-class classification problem.

The project applies a strict anti-leakage methodology by excluding all consumption aggregate variables from predictive modeling. Instead, it constructs 32 interpretable proxy features spanning housing quality, asset ownership, labor intensity, household composition, vulnerability exposure, and geographic characteristics. The system further includes region-specific models for all 11 Ethiopian administrative regions with sufficient sample sizes, as well as interactive regional wealth ranking and household prediction tools delivered through a CRISP-DM structured workflow.

1. INTRODUCTION

Poverty measurement and household welfare classification remain central challenges in development economics, particularly in Sub-Saharan Africa, where survey-based consumption data are collected infrequently and often become available only after significant delays.

Ethiopia, one of the world's most populous low-income countries, has experienced substantial structural transformation since 2000, transitioning from chronic food insecurity toward one of Africa's fastest-growing economies while simultaneously facing severe shocks, including the 2015–2016 drought, the COVID-19 pandemic, and the Tigray conflict.

Understanding the spatial and temporal distribution of household wealth within this rapidly changing environment is essential for evidence-based policymaking, social protection planning, and resource allocation.

This project addresses the challenge of predicting household consumption expenditure quintiles using observable household characteristics that can be collected without conducting full consumption diary surveys. Using five waves of LSMS-ISA / ESS survey data from 2011–2022, the system implements a leakage-free machine learning pipeline that predicts household welfare status using only legitimate proxy variables rather than direct consumption aggregates.

The practical outcome of the project is a publicly accessible Streamlit web application that enables users to:

- Explore survey datasets,
- Conduct exploratory data analysis,
- Train and compare machine learning classifiers,
- Analyze regional wealth disparities, and
- Generate household-level wealth predictions.

The platform is designed for both academic research and policy-oriented analysis.

1.1 Objectives

The primary objectives of this project are:

1. Develop a leakage-free multi-wave survey pipeline integrating heterogeneous ESS datasets into a unified analysis-ready dataset.
2. Engineer interpretable and economically meaningful proxy features from housing, demographic, labor, geographic, and vulnerability indicators.
3. Train and evaluate multiple machine learning classifiers for five-class wealth quintile prediction.
4. Construct region-specific wealth prediction models for Ethiopian administrative regions.
5. Deploy an interactive Streamlit dashboard suitable for academic and policy audiences.

1.2 Scope

The project utilizes approximately 18,500 household observations collected across five ESS survey waves between 2011 and 2022. The target variable, `cons_quint`, represents nationally defined household consumption expenditure quintiles ranging from Q1 (poorest) to Q5 (wealthiest).

The modeling framework covers Ethiopia's 11 administrative regions with sufficient sample sizes for region-specific analysis.

2. BACKGROUND AND RELATED WORK

2.1 LSMS-ISA and Ethiopian Socioeconomic Survey

The Living Standards Measurement Study – Integrated Surveys on Agriculture (LSMS-ISA), managed by the World Bank, provides nationally representative household survey microdata across Sub-Saharan Africa. Ethiopia’s implementation, known as the Ethiopian Socioeconomic Survey (ESS), is conducted collaboratively by the Central Statistical Agency (CSA) and the World Bank.

Five survey waves spanning 2011–2022 capture diverse economic conditions, including periods of rapid growth, drought, pandemic disruption, and conflict.

Each wave collects detailed information on:

- Household demographics,
- Labor and employment,
- Enterprise activities,
- Housing quality,
- Asset ownership,
- Shock exposure, and
- Consumption aggregates.

2.2 Wealth Prediction in Developing Country Contexts

Predicting household welfare using observable proxy variables is widely established in development economics. Traditional approaches include the Demographic and Health Survey (DHS) Wealth Index methodology and PCA-based asset indices.

More recently, machine learning methods have been applied to both satellite imagery and survey microdata for poverty prediction. However, existing Ethiopian poverty studies primarily focus on direct consumption aggregate construction rather than predictive classification using proxy indicators.

The Ethiopian context presents unique modeling challenges due to:

- Strong rural–urban disparities,

- Regional heterogeneity,
- Climatic shocks, and
- Temporal non-stationarity across survey waves.

2.3 Anti-Leakage Design

A major methodological contribution of this project is its strict anti-leakage framework.

Consumption aggregate variables such as:

- *food_cons_ann*,
- *nonfood_cons_ann*,
- *educ_cons_ann*,
- *total_cons_ann*,
- *nom_totcons_aeq*, and
- *spat_totcons_aeq*

are explicitly excluded from all predictive modeling because they directly determine the target variable (*cons_quint*).

Instead, the system relies solely on independently observable proxy indicators, ensuring that evaluation metrics reflect genuine predictive capability rather than artificial leakage.

3. DATA SOURCES AND DESCRIPTION

3.1 Survey Waves and Temporal Coverage

Wave	Survey Year	Format	Economic Context
W1	2011–12	CSV	Pre-MDG baseline
W2	2013–14	SPSS (.sav)	High GDP growth
W3	2015–16	CSV	El Niño drought
W4	2018–19	CSV	Pre-COVID baseline
W5	2021–22	CSV	Post-COVID and conflict

3.2 Survey Modules

The project integrates multiple survey modules:

Module	Description
<i>cons_agg</i>	Household size, survey weights, target variable
<i>sect1</i>	Household roster and demographics
<i>sect3</i>	Labor and employment
<i>sect7</i>	Non-farm enterprise activities
<i>sect8/9</i>	Housing quality and shocks

Wave-specific differences in section numbering and module availability are managed centrally within the project configuration system.

3.3 Target Variable

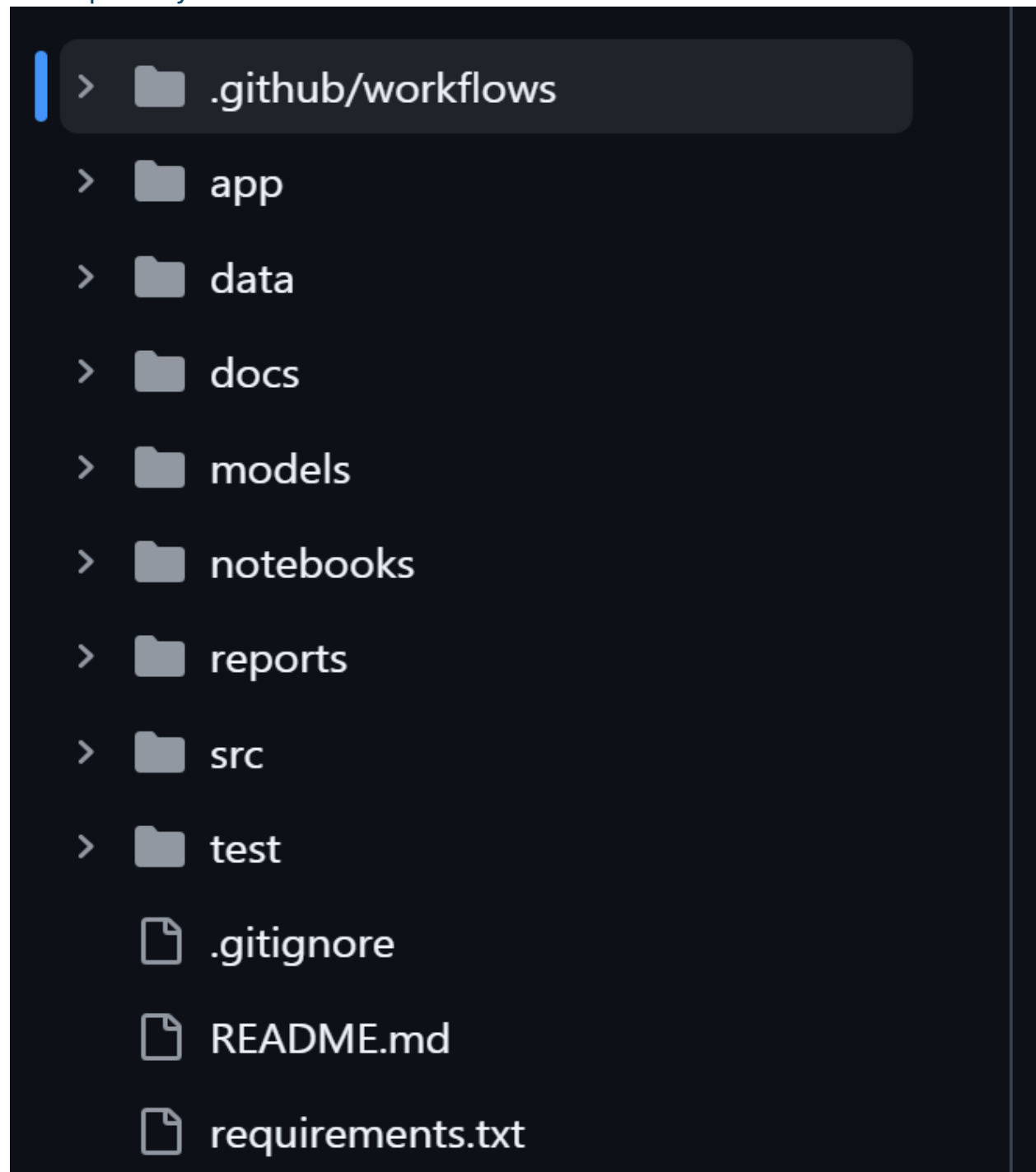
The target variable, *cons_quint*, represents household consumption expenditure quintiles ranging from:

- Q1 = poorest 20%
- Q5 = wealthiest 20%

Because the classes are balanced by construction, classification accuracy is directly interpretable relative to the 20% random baseline.

4. SYSTEM ARCHITECTURE AND DESIGN

4.1 Repository Structure



4.2 Design Principles

The system follows six key principles:

1. Leakage prevention
2. Wave-aware processing
3. Modular architecture
4. Reproducibility
5. Interpretability
6. Deployment readiness

4.3 CRISP-DM Alignment

The project follows the CRISP-DM framework:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

5. DATA PIPELINE

5.1 Data Loading and Format Normalization

The pipeline handles heterogeneous survey formats across waves, including custom SPSS decoding for Wave 2.

Key preprocessing operations include:

- Variable renaming,
- Binary variable normalization,
- Survey harmonization, and
- Leakage-safe feature selection.

5.2 Missing Value Handling

The project implements multiple imputation strategies:

- Median imputation,
- Mode imputation,
- Constant filling,
- KNN imputation,
- Iterative MICE imputation,
- Group-wise regional imputation,
- Missingness indicators.

All operations are logged for reproducibility and auditing.

5.3 Data Cleaning

Cleaning procedures include:

- Outlier capping,
- Coverage flag generation,
- Near-zero variance filtering.

5.4 Feature Engineering

The system constructs 32 interpretable features grouped into:

1. Household composition,
2. Housing quality,
3. Asset wealth,
4. Labor intensity,
5. Vulnerability exposure,
6. Geographic context,
7. Human capital.

The `housing_quality_idx` and `modern_asset_score` emerge as the most predictive features.

5.5 Preprocessing and Data Splits

The preprocessing pipeline applies:

- Median imputation for continuous variables,
- Mode imputation for categorical variables,
- Standard scaling,
- One-hot encoding.

The dataset is split into:

- Training set (65%),
- Validation set (15%),
- Test set (20%).

All splits are stratified by quintile.

6. MACHINE LEARNING METHODOLOGY

6.1 Problem Formulation

The task is formulated as a five-class household wealth classification problem:

{1,2,3,4,5}

where:

- 1 = poorest quintile,
- 5 = wealthiest quintile.

6.2 Classifier Suite

The project evaluates:

- Logistic Regression,
- Decision Tree,
- Random Forest,
- K-Nearest Neighbors,
- Naive Bayes,
- Gradient Boosting,
- AdaBoost,
- XGBoost,
- LightGBM.

Random Forest achieved the best overall performance.

6.3 Evaluation Metrics

Primary metrics include:

- Accuracy,
- Weighted F1-score,
- Macro F1-score,
- ROC-AUC,

- Confusion Matrix,
- MAE,
- RMSE.

6.4 Cross-Validation Strategy

The project uses:

- Stratified 5-fold cross-validation,
- GridSearchCV hyperparameter tuning,
- Fixed random seed (RANDOM_STATE = 42).

6.5 Per-Region Modeling

Separate regional models are trained for Ethiopian administrative regions with sufficient sample sizes, enabling regional wealth ranking and comparative analysis.

7. RESULTS AND DISCUSSION

7.1 Overall Model Performance

Metric	Value
Test Accuracy	40.81%
Weighted F1-score	0.3992
Macro F1-score	0.3714
CV Mean F1-score	0.3839
CV Std Dev	± 0.0057
MAE	1.005
RMSE	1.474

The model substantially outperforms the 20% random baseline while maintaining stable cross-validation performance.

7.2 Model Comparison

Performance ranking:

Random Forest > Gradient Boosting > XGBoost > LightGBM > AdaBoost > Logistic Regression > KNN > Decision Tree > Naive Bayes

Tree-based ensemble methods consistently performed best on structured survey data.

7.3 Feature Importance

Most important features include:

1. housing_quality_idx
2. modern_asset_score
3. has_electricity
4. region_ADDIS_ABABA
5. labour_intensity

6. shock_breadth
7. head_edu_level

7.4 Regional Wealth Analysis

Addis Ababa consistently ranks highest in predicted wealth levels, followed by Dire Dawa and Harari, while Somali, Afar, Gambela, and Benishangul-Gumuz generally rank lower.

Wave 5 results reveal notable welfare disruptions in Tigray consistent with the regional conflict period.

8. APPLICATION ARCHITECTURE

The Streamlit dashboard includes:

- Home page,
- Data Explorer,
- EDA tools,
- Preprocessing Audit,
- Modeling interface,
- Regional Wealth Map,
- Household Prediction form,
- About page.

The application is publicly deployed at: ethiowealthpredictor.streamlit.app

9. TESTING AND QUALITY ASSURANCE

The project includes automated testing covering:

- Data pipeline validation,
- Model loading,
- Prediction integrity,
- Anti-leakage enforcement,
- CI/CD via GitHub Actions.

10. ETHICAL CONSIDERATIONS AND LIMITATIONS

The system is intended strictly for:

- Academic research,
- Policy analysis,
- Educational exploration.

It is not designed for operational targeting or individual eligibility determination.

Key limitations include:

- Cross-sectional data structure,
- Temporal validity constraints,
- Missing modules in later waves,
- Unordered classification framework,
- Small regional sample sizes.

11. CONCLUSION AND FUTURE WORK

This project delivers a reproducible and deployable machine learning framework for Ethiopian household wealth prediction using nationally representative survey data.

Future improvements include:

- Ordinal regression methods,
- Deep learning for tabular data,
- Longitudinal panel modeling,
- Fairness auditing,
- Satellite imagery integration,
- Expanded survey module utilization.

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